

Assignment 1

Due date: Monday, September 28th, 12:30 pm.

Exercise 1

Consider a non-dividend-paying stock with current value $S = 100$. In the risk-neutral measure the stock follows a jump-diffusion process

$$\frac{dS}{S} = (r - \lambda^Q \gamma)dt + \sigma dW_t + \gamma dN_t \quad (1)$$

Take $\gamma = -0.08$ (i.e. the stock price is subject to downward 8% jumps), $\lambda^Q = 0.2$ (i.e. on average the stock experiences a jump every 5 years), $\sigma = 0.2$, $r = 0.04$. Compute the value of a call option with expirations $T = 0.02, 0.08, 0.25, 0.5$ (i.e. approximately 1 week, 1 month, 3 months, 6 months). and strikes $K/S = 0.8, 0.9, 1, 1.1$.

For every expiration, plot the Black-Scholes implied volatility of the option as a function of strike. Submit your code, plots and short write-up explaining what you did.